

Grade 8
Unit 3
Lesson 6 Practice

Name Answer Key
Date _____
Period _____

1. Dominic multiplied $(6.8 \times 10^4)(9.3 \times 10^4)$. His work is shown in the table. He did the problem first using standard notation and then checked his work using scientific notation. He noticed the answers didn't match.

Standard Notation	Scientific Notation
$68,000 \times 93,000$	$(6.8 \times 10^4)(9.3 \times 10^4)$
$63.24 \times 100,000,000$	$(6.8 \times 9.3)(10^4 \times 10^4)$
$6,324,000,000$	6.324×10^8

Find Dominic's mistake and make the correction.

$$6.8 \times 9.3 = 63.24 \qquad 63.24 \times 10^8 = 6.324 \times 10^9$$

2. Complete the work to find the product. Be sure to write your answer in scientific notation.

$$\begin{aligned} &(7.1 \times 10^3)(8.7 \times 10^5) \\ &= \underline{(7.1 \times 8.7)}(10^3 \cdot 10^5) \\ &= 61.77 \times 10^8 \\ &= \underline{6.177} \times 10^9 \\ &\quad 6.177 \times 10^9 \end{aligned}$$

3. Find the product of $(8.4 \times 10^9)(8.4 \times 10^{11})$.

$$\begin{aligned} &(8.4 \times 8.4)(10^9 \times 10^{11}) \\ &70.56 \times 10^{20} \\ &7.056 \times 10^{21} \end{aligned}$$

4. Find the product of $(1.44 \times 10^{-8})(6.0 \times 10^{-5})$.

$$\begin{aligned} &(1.44 \times 6)(10^{-8} \times 10^{-5}) \\ &8.64 \times 10^{-13} \end{aligned}$$

5. Find the product of $(3 \times 10^4)(9.9 \times 10^7)$.

$$\begin{aligned} &(3 \times 9.9)(10^4 \times 10^7) \\ &29.7 \times 10^{11} \\ &2.97 \times 10^{12} \end{aligned}$$

Match each product with its expression in scientific notation using the answer bank for questions 6-10. Write the correct equivalent expression next to each product.

3.6×10^{15}	3.212×10^9
5.76×10^{-5}	1.8×10^{-15}
4.539×10^7	

6. $(7.3 \times 10^2)(4.4 \times 10^6)$

3.212×10^9

7. $(2.5 \times 10^{-9})(7.2 \times 10^{-7})$

1.8×10^{-15}

8. $(8.9 \times 10^2)(5.1 \times 10^4)$

4.539×10^7

9. $(2.4 \times 10^{-2})(2.4 \times 10^{-3})$

5.76×10^{-5}

10. $(8 \times 10^8)(4.5 \times 10^6)$

3.6×10^{15}